

CHAPTER 1 — ACCESSING DIGITS

SLIDE 1 — WHAT IS NUMBER CRUNCHING?

MEET NUMBER CRUNCHING!

We normally work with a complete number.

But sometimes, we need to work with the individual digits of a number.

Simple Definition

Number crunching means working with the individual digits of a number.

LOOK AT THIS!

"1234"

1 2 3 4



Each one is a digit.

Together → Number 

EASY ANALOGY

Think of a number like a row of boxes 



1 2 3 4

Each box contains one digit.

REMEMBER!

****One digit →

1	2
3	4

****Many digits together → Number****

SLIDE 2 — ACCESSING DIGITS

HOW DO WE ACCESS A DIGIT?

A number contains many digits.

For example:

1234

Sometimes, we need to **access each digit separately**.

Simple Definition

Accessing digits means getting the individual digits from a number one by one.

LOOK AT THIS!

1234

We can access:

4 → 3 → 2 → 1

We start from the **last digit**.

HOW?

In C, we use the **modulus operator %**.

`digit = n % 10;`

For example:

`n = 1234`

`1234 % 10 = 4`

So:

Last digit = 4

EASY ANALOGY

Think of a number like a row of boxes 

 1 →  2 →  3 →  4



Take this first!

We access the **last digit first**.

SLIDE 3 — REMOVING THE LAST DIGIT

HOW DO WE REMOVE A DIGIT?

After accessing the last digit, we need to **remove it**.

In C, we use **division /**.

REMOVE THE LAST DIGIT

`n = n / 10;`

For example:

`1234 / 10 = 123`

So:

$1234 \rightarrow 123$

The last digit 4 is removed.

 **LOOK AT THIS!**

$1234 \rightarrow 123$

$123 \rightarrow 12$

$12 \rightarrow 1$

$1 \rightarrow 0$

Each division by 10 removes **one digit**.

 **EASY ANALOGY**

Think of a number like a row of boxes 

 1 $\rightarrow$  2 $\rightarrow$  3 $\rightarrow$  4

Remove the last box:

 1 $\rightarrow$  2 $\rightarrow$  3

$1234 \rightarrow 123$

★ **REMEMBER!**

/ 10 $\rightarrow$ Removes the last digit $\div$

 **SLIDE 4 —  ACCESSING ALL DIGITS**

 **CAN WE ACCESS ALL THE DIGITS?**

Yes! 

We can use:

% 10 → Get the last digit

/ 10 → Remove the last digit

Then repeat the process.

 EXAMPLE

Let's take:

1234

STEP 1

$$1234 \% 10 = 4$$

$$1234 / 10 = 123$$

👉 Digit = 4

STEP 2

$$123 \% 10 = 3$$

$$123 / 10 = 12$$

👉 Digit = 3

STEP 3

$$12 \% 10 = 2$$

$$12 / 10 = 1$$

👉 Digit = 2

STEP 4

$$1 \% 10 = 1$$

$$1 / 10 = 0$$

👉 Digit = 1

 **SO WE GET:**

1234

↓

4 → 3 → 2 → 1

 **BIG MEMORY TRICK**

% 10 → GET 

/ 10 → REMOVE 

 **TAKEAWAY**

Use % 10 and / 10 together to access digits one by one.

Questions

1 Get the Last Digit

```
int n = 1234;
```

```
int digit;
```

```
digit = n ____ 10;
```

```
printf("%d", digit);
```

 Hint: Which operator gives the remainder?

 Glossary:

% → Modulus operator → gives the remainder

2 Remove the Last Digit

```
int n = 1234;
```

```
n = n ____ 10;
```

```
printf("%d", n);
```



Hint: Which operator removes the last digit?



Glossary:

/ → Division operator → removes the last digit when dividing an integer by 10.

③ What Happens Next?

```
int n = 5678;
```

```
int digit = n % 10;
```

```
n = n / 10;
```

What are the values of digit and n?

A) digit = 8, n = 567

B) digit = 7, n = 568

C) digit = 8, n = 5678

D) digit = 7, n = 567



Hint: First GET, then REMOVE.



Glossary:

GET → Access the last digit

REMOVE → Remove the last digit

④ Trace the Digits

If:

```
int n = 1234;
```

and we repeatedly use % 10 and / 10, what order will the digits be accessed?

- A) $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$
- B) $4 \rightarrow 3 \rightarrow 2 \rightarrow 1$
- C) $1 \rightarrow 4 \rightarrow 2 \rightarrow 3$
- D) $4 \rightarrow 1 \rightarrow 3 \rightarrow 2$

 Hint: % 10 always starts from the right side.

 Glossary:

Access $\rightarrow$ To get or retrieve a value.

5 If $n = 5678$, which digit is accessed first?

- A) 5
- B) 6
- C) 7
- D) 8

 Hint: % 10 starts from the right side.

 Glossary:

Access $\rightarrow$ To get or retrieve a value.